

Andres Perez

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EDUCATION

University of Notre Dame | Notre Dame, IN

Graduation Date: May 2028

Bachelor of Science: Computer Engineering | Minor: Engineering Corporate Practice

GPA: 3.713

Engineering Study Abroad: London, England

May 2025 – June 2025

Relevant Courses: Data Structures, Discrete Math, Intro. to Electrical and Computer Engineering, Intro. To Embedded Systems

ENGINEERING LEADERSHIP

IrishSat's Gravitational Orbital Attitude Thermal Lab (GOAT Lab)

Notre Dame, IN

Electronics Lead | CubeSat Team

Aug. 2024 – Present

- Spearheading a team of 6+ engineers for the embedded systems development and PCB design initiatives for testing equipment, particularly a Helmholtz cage project, coordinating hardware-software integration across CubeSat projects.
- Collaborated under expert mentorship to refine IrishSat's Helmholtz Cage design, integrating PCB design (KiCad), control algorithms (MATLAB), and embedded Linux systems (Raspberry Pi) into a unified testing platform.
- Designed and assembled a custom 7.2V 7000 mAh 2S2P Li-ion battery pack using 18650 cells and spot welding.

ENGINEERING PROJECTS

Hardware-Accelerated Human Activity Classifier | Independent Project

Long Beach, CA

Designer

July 2025 – Aug. 2025

- Built a wireless data acquisition system using two Arduino Uno R3s, an MPU-9250 IMU sensor, and nRF24L01 RF transceivers.
- Developed a C++ sketch to stream accelerometer/gyroscope data wirelessly and a Python script to parse it into a labeled walking/sitting dataset.
- Achieved 96.7% classification accuracy with a PyTorch 1D CNN, documenting performance across various hyperparameters.

Helmholtz Cage Driver PCB | GOAT Lab

Notre Dame, IN

Lead Designer

Mar. 2025

- Designed a custom Arduino R3 hat PCB using KiCad to integrate and optimize Arduino, magnetometers, and H-bridge circuits, reducing assembly complexity.
- Amended the generated Bill of Materials and Component Placement List files export to JLCPCB for manufacturing.

Air Quality Sensor Housing | Engineering Design

Notre Dame, IN

Lead Designer

Dec. 2024

- Collaborated with a team of four engineering students to develop and iterate on sensor housing with a rack-and-pinion opening mechanism actuated by a servo motor and ESP32-based microcontroller.
- Designed and prototyped with SolidWorks in two design phases, optimizing for form, fit, function, and durability through iterative testing and feedback.

Portfolio Website | Independent Project | andres-perez.github.io

Long Beach, CA

Developer

July 2024– Present

- Developed a student and engineering project portfolio website with Astro and Tailwind CSS frameworks as the front end.

RESEARCH & WORK EXPERIENCE

Student Researcher | Notre Dame Department of Electrical Engineering

Aug. 2025 – Present

- Developing a real-time Python software interface for tracking and visualizing GPS-enabled test resources, for drones and software-defined radios, that parses NMEA strings to extract GPS data and displays it on a live map using the folium library, supporting various research experiments.

Engineering Career Assistant | Meruelo Family Center for Career Development

Sept. 2025 – Present

- Coauthoring a bi-weekly newsletter for undergraduates, specifically seniors, highlighting career opportunities and alumni spotlights from various majors and post-graduate experiences.
- Grew the newsletter to 76.3% open rate with a 2.3% click rate of 517 recipients.

TECHNICAL AND LANGUAGE SKILLS

Skills: Microsoft Office Suite (Excel, PowerPoint, Word), KiCad, Vim, SolidWorks, Technical Writing, Electronics Test Bench (Function Generator, Oscilloscope, Multimeter, LCR meter, Power Supplies, Logic Analyzer, Electronics Soldering)

Programming: C/C++, Python (Threading, Socket, PyTorch, Pandas, NumPy), Git, GitHub, MATLAB, Astro

Language: English (Native), Spanish (Native), Japanese (Elementary)

Interests: Embedded Systems, Robotics, Machine Learning, Data Acquisition, Avionics, PCB Design